

The Lévy Distribution

Introduction

The Lévy distribution is a continuous probability distribution named after the French mathematician Paul Lévy. It is characterized by its heavy right tail and is a special case of the stable distributions family. The Lévy distribution is particularly useful for modeling extreme value phenomena and has important applications in finance, physics, and signal processing.

Mathematical Definition

The probability density function of a Lévy distribution is given by:

$$f(x; \mu, \sigma) = \sqrt{\frac{\sigma}{2\pi}} \frac{\exp\left(-\frac{\sigma}{2(x-\mu)}\right)}{(x-\mu)^{3/2}}$$

where $x > \mu$, μ is the location parameter, and $\sigma > 0$ is the scale parameter.

Properties

The Lévy distribution possesses several distinctive properties. Unlike many common distributions, the Lévy distribution does not have a defined mean or variance, as both integrals diverge. This heavy-tailed behavior makes it particularly suitable for modeling extreme events and rare occurrences.

The Lévy distribution is closely related to the Brownian motion and appears naturally in the study of first-passage times. It is also a stable distribution, meaning that the sum of independent Lévy-distributed random variables follows the same distribution. Due to these properties, the Lévy distribution is extensively used in financial mathematics for modeling asset price movements and in physics for studying diffusion processes.